

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 123 — Past Questions & Materials
by Renance

Course: MTH 123
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 100

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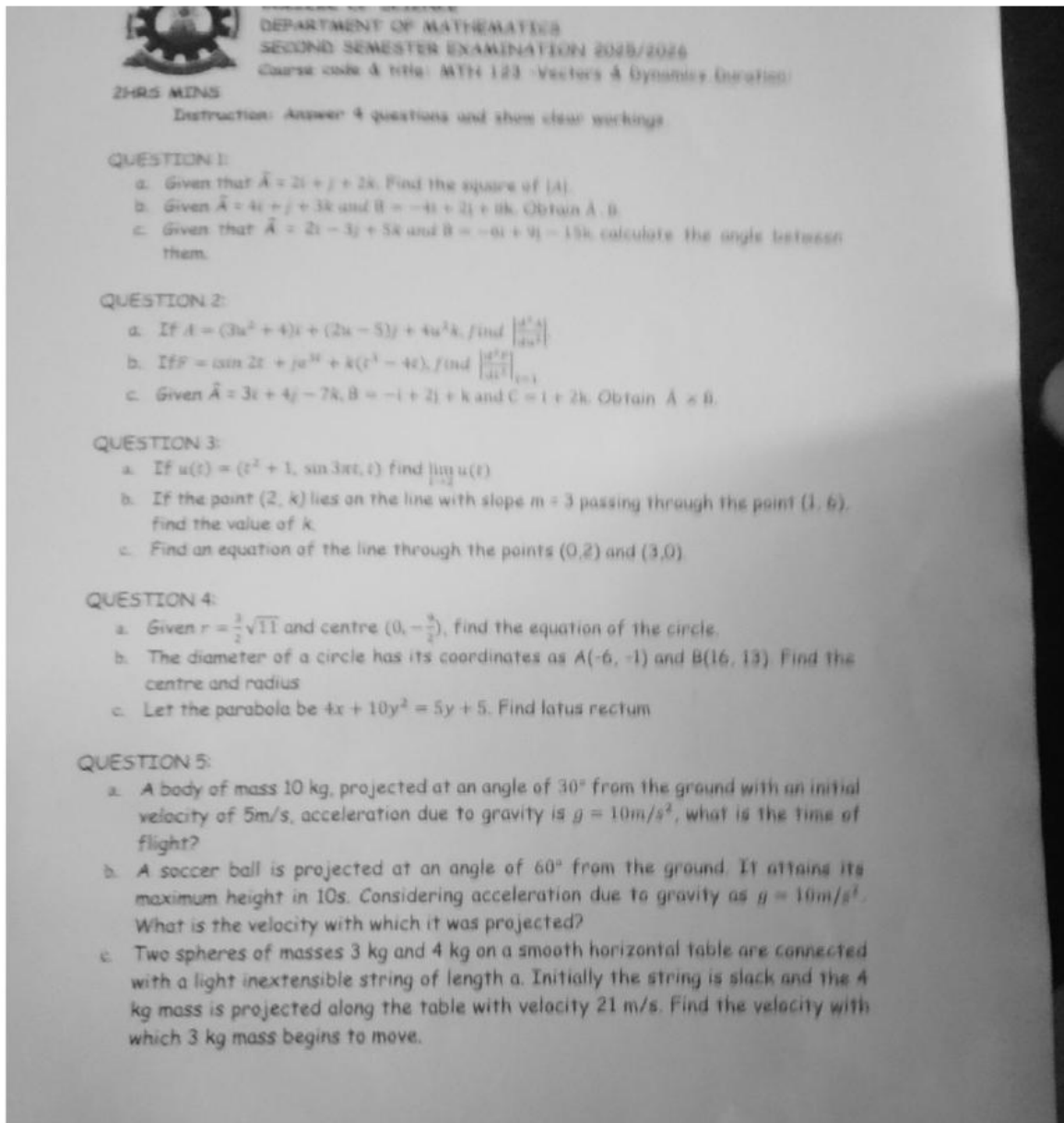
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Paper 1 of 1 — MTH 123/107 VECTOR AND DYNAMICS 2025/2026

Level 100 • First Semester • PastQuestion

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DEPARTMENT OF MATHEMATICS
SECOND SEMESTER EXAMINATION 2025/2026
 Course code & title: MTH 123 -Vectors & Dynamics (duration: 2HRS MINS)

Instruction: Answer 4 questions and show clear workings

QUESTION 1:

- Given that $\vec{A} = 2i + j + 2k$. Find the square of $|\vec{A}|$.
- Given $\vec{A} = 4i + j + 3k$ and $\vec{B} = -4i + 2j + 8k$. Obtain $\vec{A} \cdot \vec{B}$.
- Given that $\vec{A} = 2i - 3j + 5k$ and $\vec{B} = -6i + 9j - 15k$ calculate the angle between them.

QUESTION 2:

- If $\vec{A} = (3u^2 + 4)i + (2u - 5)j + 4u^2k$, find $\left| \frac{d\vec{A}}{du} \right|$.
- If $\vec{r} = i \sin 2t + j e^{3t} + k(t^3 - 4t)$, find $\left| \frac{d^2\vec{r}}{dt^2} \right|_{t=1}$.
- Given $\vec{A} = 3i + 4j - 7k$, $\vec{B} = -i + 2j + k$ and $\vec{C} = i + 2k$. Obtain $\vec{A} \times \vec{B}$.

QUESTION 3:

- If $u(t) = (t^2 + 1, \sin 3\pi t, t)$ find $\lim_{t \rightarrow 2} u(t)$.
- If the point $(2, k)$ lies on the line with slope $m = 3$ passing through the point $(1, 6)$, find the value of k .
- Find an equation of the line through the points $(0, 2)$ and $(3, 0)$.

QUESTION 4:

- Given $r = \frac{3}{2}\sqrt{11}$ and centre $(0, -\frac{9}{2})$, find the equation of the circle.
- The diameter of a circle has its coordinates as $A(-6, -1)$ and $B(16, 13)$. Find the centre and radius.
- Let the parabola be $4x + 10y^2 = 5y + 5$. Find latus rectum.

QUESTION 5:

- A body of mass 10 kg, projected at an angle of 30° from the ground with an initial velocity of 5m/s, acceleration due to gravity is $g = 10\text{m/s}^2$, what is the time of flight?
- A soccer ball is projected at an angle of 60° from the ground. It attains its maximum height in 10s. Considering acceleration due to gravity as $g = 10\text{m/s}^2$. What is the velocity with which it was projected?
- Two spheres of masses 3 kg and 4 kg on a smooth horizontal table are connected with a light inextensible string of length a . Initially the string is slack and the 4 kg mass is projected along the table with velocity 21 m/s. Find the velocity with which 3 kg mass begins to move.

